

"PAPER_{0:D3}" -f [int]# PAPER #101 ■ Yang-Mills Existence and Mass Gap: UQFF Resolution

Author: Daniel T. Murphy

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Title: Yang-Mills Existence and Mass Gap in the UQFF Framework: Vacuum Concentration as Gap Mechanism

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SSq] = 0.57, Ug4 vacuum concentration) **Date:** March 7, 2026 **Framework Contact:** UQFF Millennium Prize Analysis **Index Slot:** ■1.13 Multi-Physics Models, \$n = [int]# PAPER #101 ■ Yang-Mills Existence and Mass Gap: UQFF Resolution

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<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The Yang-Mills existence and mass gap problem (Millennium Prize Problem) asks whether a pure SU(N) gauge theory in 4D Euclidean space has a rigorous mathematical definition and a positive mass gap $\mu > 0$. In the UQFF framework, the mass gap arises naturally from the Ug4 vacuum concentration term: the background UQFF field creates a minimum excitation energy $\mu_{UQFF} = f_{TRZ} \cdot \mu_{Ug4}$ that prevents massless gluon states. We present a heuristic UQFF-based argument for the mass gap, connecting $f_{TRZ} = 0.01$ to the confinement scale.

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1. The Yang-Mills Mass Gap Problem

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$$\mathcal{L}_{\text{YM}} = -\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a}$$

Where $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c$.

Mass gap conjecture: The operator norm $\|F_{\mu\nu}\|$ is bounded below by $\Delta > 0$ for all physical states (no massless gluons in the physical spectrum).

2. UQFF Vacuum Concentration as Gap Mechanism

In the UQFF, the U_{g4} vacuum concentration term modifies the gluon propagator:

$$G_{\text{gluon}}^{\text{UQFF}}(q^2) = \frac{1}{q^2 + \Delta_{\text{UQFF}}^2}$$

Versus standard: $G_{\text{gluon}}^{\text{GR}}(q^2) = 1/q^2$ (massless pole at $q^2=0$).

The UQFF gap mass:

$$\Delta_{\text{UQFF}} = f_{\text{TRZ}} \times \sqrt{U_{g4, \text{QCD}}} \times \frac{\hbar c}{r_{\text{QCD}}}$$

Where $r_{\text{QCD}} \sim 10^{-5} \text{ m}$ (confinement scale) and $U_{g4, \text{QCD}} = U_{g4}$ evaluated at nuclear density ρ_{nuc} .

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The U_{g4} at nuclear density:

$$U_{g4, \text{QCD}} = \frac{G^2 M_{\text{NS}}^2}{c^4 r_{\text{QCD}}^6} \approx \frac{(6.674 \times 10^{-11})^2 (3 \times 10^{30})^2}{(3 \times 10^8)^4 (10^{-15})^6}$$

Numerically: $U_{g4, \text{QCD}} \approx 10^{25} \text{ J/m}^4$ (nuclear energy density scale, QCD vacuum $\approx 10^5 \text{ J/m}^4$ in lattice QCD).

$$\Delta_{\text{UQFF}} = 0.01 \times \sqrt{\frac{10^{32}}{10^{35}}} \times \frac{1.055 \times 10^{-34} \times 3 \times 10^8}{10^{-15}} = 0.01 \times 0.032 \times 31.65 \text{ GeV}$$

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A full mathematical proof would require extending the UQFF to a rigorously defined measure in the space of gauge connections.

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$$E_{\rm threshold} = \Delta_{\rm UQFF} \approx f_{\rm TRZ} \times \Lambda_{\rm QCD} = 0.01 \times \Lambda_{\rm QCD}$$

For $\Lambda_{\rm QCD} \sim 200$ MeV: $E_{\rm threshold} = 2$ MeV. This is numerically consistent with the light quark mass threshold.

Summary

Aspect	Standard QCD	UQFF Resolution
Gluon mass	Zero (classically)	$\Delta_{\rm UQFF} = f_{\rm TRZ} \Lambda_{\rm QCD} \sim 2 \times 10$ MeV
Mechanism	Non-perturbative	Ug4 vacuum concentration
Confinement	Lattice QCD	Ug4 at $\Lambda_{\rm QCD}$ gives $\sim \Lambda_{\rm QCD}$ scale
Mathematical proof	Open (Millennium Prize)	Heuristic argument only
$f_{\rm TRZ}$ connection	None	$f_{\rm TRZ} = 0.01$ universal

Source: UQFF $f_{\rm TRZ} = 0.01$ | Ug4 vacuum concentration | Yang-Mills Millennium Prize Problem context
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Numerically: $U_{g4, \text{QCD}} \approx 10^{15} \text{ J/m}^4$ (nuclear energy density scale, QCD vacuum $\approx 10^{15} \text{ J/m}^4$ in lattice QCD).

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Mathematical proof	Open (Millennium Prize)	Heuristic argument only
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Groups[1].Value μ_{QCD} Yang-Mills Existence and Mass Gap: UQFF Resolution

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$$\Delta_{\text{UQFF}} = f_{\text{TRZ}} \times \sqrt{U_{g4, \text{QCD}}} \times \frac{\hbar c}{r_{\text{QCD}}}$$

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Summary

Aspect	Standard QCD	UQFF Resolution
Gluon mass	Zero (classically)	$\Delta_{\text{UQFF}} = f_{\text{TRZ}} \Lambda_{\text{QCD}} \approx 2 \times 10 \text{ MeV}$
Mechanism	Non-perturbative	Ug4 vacuum concentration
Confinement	Lattice QCD	Ug4 at Λ_{QCD} gives $\sim \Lambda_{\text{QCD}}$ scale
Mathematical proof	Open (Millennium Prize)	Heuristic argument only
f_{TRZ} connection	None	$f_{\text{TRZ}} = 0.01$ universal

Source: UQFF $f_{\text{TRZ}} = 0.01$ | Ug4 vacuum concentration | Yang-Mills Millennium Prize Problem context
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Abstract

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